switch sense circuits, from the resistor multiplexed panel lamps dimmer circuitry of the left multi-function switch on the panel lamps dimmer signal circuit and its control logic to provide a battery current output to the courtesy lamps on a courtesy lamp feed circuit. The EMIC control provides a theater-type fade-to-off feature that will slowly dim the courtesy lamps about five seconds after both doors are closed.

The EMIC interior lighting control programming provides an illuminated entry/exit feature by monitoring the door ajar and ignition switch inputs. When a door is opened with the ignition switch in the Off position, the EMIC turns on the courtesy lamps. When the ignition switch is turned to the On or Start positions, the EMIC turns the courtesy lamps off immediately with no theater dimming. When the ignition switch is turned from the On position to the Off position, the EMIC turns on the interior lights for about ten seconds or until the ignition switch is again turned to the On or Start positions, whichever occurs first.

The EMIC interior lighting control programming also provides a battery saver feature (load shedding) for the interior lighting. Unless the engine is running, the EMIC will automatically turn off the interior lights if they are left on for more than about twenty minutes, regardless of the status of the ignition switch, door ajar switch, or left multi-function switch inputs to the cluster.

The hard wired inputs and output of the EMIC interior lighting control may be diagnosed using conventional diagnostic methods; however, there are no other diagnostic tools available for the EMIC timer and logic circuitry. If the input and output components and circuits of the interior lighting system test OK, but the system fails to operate as designed, the EMIC must be replaced.

REAR WINDOW DEFOGGER CONTROL

The EMIC contains an integral timer and logic circuit to perform the rear window defogger timer and control functions for the optional rear window defogger system. The EMIC uses a hard wired input from the rear window defogger switch on the rear window defogger switch sense circuit and its control logic to determine the correct output to the rear window defogger relay. The EMIC controls the ground path of the rear window defogger relay control coil through an output on the rear window defogger relay control circuit.

The EMIC is programmed to interpret each momentary ground signal it receives on the rear window defogger switch sense circuit as a request to change the current state of the output on the rear window defogger relay control circuit. Therefore, with the ignition switch in the On position, the first ground input on the rear window defogger switch sense circuit turns the system On, the second ground input turns the system Off, and so forth. Once the rear window defogger system has been turned On, it can be turned off manually by depressing the rear window defogger switch a second time or by turning the ignition switch to the Off position.

The timer function of the EMIC will also automatically turn the rear window defogger system Off. The timer turns the system Off after about ten minutes of operation; however, after the first timed interval has expired, each time the system is turned On again during that same ignition cycle, the timer will automatically turn it Off after about five minutes of operation.

The hard wired input and output of the EMIC rear window defogger control may be diagnosed using conventional diagnostic methods; however, there are no other diagnostic tools available for the EMIC rear window defogger timer and logic circuitry. If the input and output components and circuits of the rear window defogger system test OK, but the system fails to operate as designed, the EMIC must be replaced.

INPUT AND OUTPUT CIRCUITS

HARD WIRED INPUTS

The hard wired inputs to the EMIC include the following:

- **ABS Warning Indicator Driver**
- **Brake Warning Indicator Sense**
- **Driver Door Ajar Switch Sense**
- **Fog Lamp Feed**
- **Front Axle Lock Request (w/Off-Road Package Only)**
- **Front Lock Indicator Switch Sense (w/Off-Road Package Only)**
- **Fused B(+) - Ignition-Off Draw**
- **Fused Ignition Switch Output (Run-Start)**
- **Headlamp Switch Output**
- **High Beam Indicator Driver**
- **Key-In Ignition Switch Sense**
- **Left Turn Signal**
- **Panel Lamps Dimmer Signal**
- **Passenger Door Ajar Switch Sense**
- **Rear Axle Lock Request (w/Off-Road Package Only)**
- **Rear Lock Indicator Switch Sense (w/Off-Road Package Only)**
- **Rear Window Defogger Switch Sense**
- **Right Turn Signal**
- **Seat Belt Switch Sense**
- **4WD Switch Sense (w/o Off-Road Package only)**

Refer to the appropriate wiring information for additional details.